

Dissolving Investigation

Year 6 Science | Australian Curriculum v9.0 | AC9S6U04 | Inquiry: AC9S6I01-I03

Name: _____ Date: _____

LEARNING INTENTION & SUCCESS CRITERIA

I can plan and carry out a fair test of dissolving time.

I can identify variables, repeat measurements and use evidence to explain my results.

QUESTION & PREDICTION

Question: Does stirring change how long sugar takes to dissolve?

Measure: time until no sugar crystals are visible. This tests dissolving speed, not the maximum amount that can dissolve.

1. I predict _____ will dissolve faster because...

PLAN A FAIR TEST

Materials: two matching clear cups, room-temperature water, white granulated sugar, measuring cylinder, balance, spoon and timer.

Each trial: 100 mL water and 1 g sugar in each cup.

Adult: pre-test the amount and observation time with your equipment.

2. What will you change? What will you measure?

3. Name three things you will keep the same.

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METHOD - READ BEFORE STARTING

1. Label matching cups A (stirred) and B (unstirred). Add 100 mL of water at the same room temperature to each.
2. Measure 1 g sugar for each cup. Test one cup at a time.
3. Add sugar and start the timer immediately.
4. For A, stir gently at about one circle each second. For B, do not stir or move the cup.
5. Stop timing when no sugar crystals are visible. View through the side and bottom without moving B.
6. Record time in seconds. Repeat with fresh water and sugar for three trials per condition.

SAFETY & OBSERVATION LIMIT

Do not taste the mixtures. Use plastic cups and wipe spills promptly.
Set a 10-minute limit. If crystals remain, record "not finished by 600 s"; do not write 600 s as the completed dissolving time.

RESULTS TABLE

Trial	Stirred: time (s)	Unstirred: time (s)
1		
2		
3		

OBSERVATION NOTES

What did you notice about the crystals? _____

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PART B - USE YOUR EVIDENCE

4. Which condition dissolved faster in your trials? Use at least two recorded times. If a trial did not finish, explain what you can still conclude.

5. Did the results support your prediction? Explain.

PART C - EVALUATE THE TEST

6. A student uses warm water for A and cold water for B. Why is this an unfair test of stirring?

7. Identify one possible source of error in your test. Suggest an improvement.

REFLECTION & NEXT QUESTION

8. Faster dissolving does not prove that more sugar can dissolve.
Write a different question about dissolving that you could test.

I can use recorded evidence in a conclusion: Yes / Not yet

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ANSWER KEY & EXPECTED REASONING

1. Accept a reasoned prediction. Stirring is normally expected to reduce dissolving time under these conditions.
2. Change: stirring / no stirring. Measure: dissolving time in seconds.
3. Any three: water volume, initial water temperature, sugar mass/type/grain size, cup size/shape. Keep the stirred condition's stirring pace consistent across trials.
4. Answers must use actual recorded results. Accept unexpected outcomes when supported by data. "Not finished by 600 s" indicates a time longer than the limit, not an exact completion time.
5. Compare the prediction with the observations; no invented results.
6. Both temperature and stirring change, so the effect of stirring cannot be isolated.
7. Examples: inconsistent stirring, delayed timer start or subjective endpoint. Improve using a set pace, same timer procedure and agreed endpoint.
8. Example: "Does water temperature change the time taken for 1 g sugar to dissolve?" Change one variable at a time.

TEACHING TIP & NEXT STEP

Pre-test before teaching. If the unstirred trial is too slow, use a smaller sugar mass consistently in every trial and record the adapted method. Split repeated trials among groups using the same method if time is limited.

Optional: calculate mean times only when all three trials have completed. Do not include an unfinished trial as 600 s. Do not replace sugar with a fizzy tablet: that introduces a chemical reaction.

TEACHER FEEDBACK

Strength: _____ Next step: _____